ELخLAAAASA technical schools 2025

A. Scientific terms

1- Magnetic field:

It is the region in which the magnetic force appears through it.

2- The magnetic flux (Φ):

It is the total number of the magnetic field lines that pass perpendicular to surface area.

3- Magnetic flux density at a point (B):

*It is the magnetic flux that passes through normal a unit area around this point

*It is the magnetic force that acts on a straight wire of length 1m , carrying electric current 1A placed normal to a magnetic field.

4- Tesla:

*A magnetic force of 1N acts on a wire carrying current 1A , placed normal to a magnetic field

* the magnetic flux lines that passes normal to a unit area = 1 wb

5- The magnetic field:

It is the region that appears the magnetic force though it

6- Magnetic permeability μ :

It is the ability of the medium to allow passing the magnetic field through it.

7- Neutral point:

*It is the point that the magnetic flux density vanishes through it ($B_t = 0$).

8. Electromagnetic induction:

It is a phenomenon of producing induced electromotive force and induced electric current in a conductor due to the variation of magnetic flux lines that is linked by a conductor

9- Lenz's rule:

The induced current that passes in a conductor opposes the variation that producing it

10- Faraday's law:

The induced electromotive force in a conductor is directly proportional with the rate of variation of magnetic flux lines linked by a conductor ($\Delta\emptyset/\Delta t$) and is directly proportional with the no.of turns of the conductor that cut the magnetic flux (N)

11- The mutual induction between two coils:

It is the electromagnetic induction that is produced in 2^{nd} coil due to the variation of electric current in 1^{st} coil. the induced current opposes the variation of the electric in 1^{st} coil.

12- The coefficient of mutual induction between coil (M):

It is the induced electromotive force in a coil due to the variation of electric current in other coil by 1A in 1sec.

13- The self-induction:

It's electromagnetic induction that produced in a coil due to the variation of electric current in same coil to oppose this variation

14- The coefficient of self induction in a coil (L):

It’s induced EMF in a coil due to the variation of electric current in the same coil by 1A in 1sec

15- Henry:

It’s is the mutual induction between 2 coil to produce EMF = 1V in a coil when , the electric current changes in other coil by rate 1 A/s.

16- Eddy currents:

They are induced currents that are generated in a metallic block as a closed path due to the variation of a magnetic flux and generate heat.

17- AC generator:

It is a device that is used to convert the kinetic energy into electric energy

18- Alternating current (AC):

*It’s electric current changes its direction and its intensity periodically

19- The transformer:

It is a device that is used to step up or step down in the voltage

20- The efficiency of electric transformer:

*It’s the ratio between the electric power in secondary coil (OUT PUT) to the electric power in primary coil (IN PUT)

21- DC motor:

It is a device that is used to convert the electric energy into kinetic energy(torque)

22- Black body:

it is the body that absorbs all radiation falling on it (perfect absorber) and reemits again (perfect emitter)

23- Plank's curve:

it is a graphical relation between the intensity of continuous radiations that emits from any hot bodies and the wavelengths of these radiations

24- Plank's distribution:

It is the ditribution between The intensity of radiations that emits from hot bodies and the wavelength.

25- Wien's law:

the wavelength (λ_m) accompanying the maximum intensity of radiation is inversely proportional with the kelvin temperature of hot bodies.

26- Night vision system:

imaging the moving objects in the dark due to the infrared radiations that emit from it.

27- Remote sensing:

it is a technology that is used to indicate the nature resources at the surface of earth by analyzing the infrared radiations of the earth.

28- Surface potential barrier:

it is the attractive force between the positive ions in the metallic surface and the free electrons

29- Thermal emission (thermionic effect):

it is the emission (liberation – escaping) of free electron from the metallic surface by heat.

30- Photoelectric effect:

it is the emission of free electros from the metallic surface by light falling with frequency is greater than the critical frequency.

31- Work function:

it is the minimum energy required to liberate free electrons from the metallic surface without gaining energy.

32- Critical frequency:

it is the minimum frequency of the incident light on a metallic surface to liberate electrons from this surface.

33- Photoelectric current:

it is the emitted electrons from the metallic surface due to falling light of frequency greater than the critical frequency.

34- Compton's effect:

when a photon of x – rays or γ rays collides with an electron at rest, its frequency decreases and its direction changes and the velocity of electron increases and its direction changes.

35- Photon:

it is a quantum of energy ($h\nu$) has no charge and has particle nature.

36- electron:

it is a particle with –ve charge and has wave nature.

37- x-rays

They are invisible electromagnetic waves of high energy and short wavelength ranges between the wavelength of gamma rays and the ultraviolet rays

38-Continuous spectrum (connected)

It is the spectrum which includes all wavelength and frequencies in a continuous range

39- line spectrum

It is the spectrum at specified frequencies and not continuously distributed

40 - Laser:

amplifying the light by stimulated emission of radiation

41- lifetime:

it is the period of time in which the excited atom remains in the excited level. around ($10^{-8} sec$).

42- Spontaneous emission:

it is the emission of radiation that happens when an excited atom relaxes to ground state after the lifetime is over.

43- Stimulated emission:

it is the emission of radiation that happens when an external photon stimulates an excited atom to relax into the ground state before the lifetime is over.

44- pumping process:

it is the process of excitation of the active medium

45- State of population inversion:

it is the state in which the no. of atoms of active medium in the excited state is greater than the no. of atoms in ground state.

46- Metastable energy level:

it is the energy level that has long lifetime around ($10^{-3} sec$).

47- Crystal:

Regular arrangement of atoms in solid state

48-Holes:

Free space left by electron in broken bonds in the semi-conductors crystal

49- Dynamic (thermal) equilibrium:

It is the state in which the number of broken bonds per a second equal to the number of mended bonds per a second in the semi-conductors crystal, so the number of free electrons is equal the number of holes at every temperature.

50- Pure semi-conductor:

Semi-conductors in which concentration of n (free electrons) = concentration of p (holes) at any temperature.

51- Impurity atom:

Atom of a trivalent element of pentavalent element added to a pure semi-conductors crystal to increase its conductivity

52- P-type semi-conductor:

it is a semi-conductor doped with a trivalent element to make the concentration of holes (p) greater than concentration of free electrons (n).

53- N-type semi-conductor:

it is a semi-conductor doped with a pentavalent element to make the concentration of free electrons (n) greater than the concentration of holes (p).

54- Diode:

Two adjacent crystals one of them is n-type and the other is p-type

55- Diffusion current:

it is the current which is resulted due to the diffusion of charge carriesrs from high concentration to low concentration.

56-Drift current:

it is the current which is resulted due to the internal electric field and it opposes to the diffusion current.

57- Potential barrier: The lowest potential in the contact area between the two crystal that prevent more diffusion

58-current division $\propto_e$:

the ratio between collector current to emitter current

59-current gainβ_e:

it is the ratio between collector current to base current

60- Analog electronics:

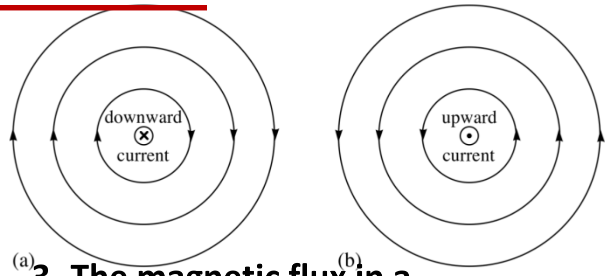
Electronics that deal with the natural quantities and convert them to continuous electrical signal

61- Digital electronics: Electronics that deal with the natural quantities and convert them to codes of zeros and one

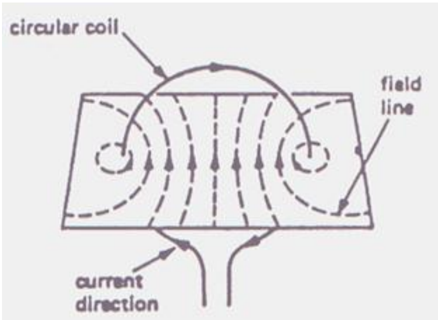
Main tablets unit one

a- Important forms

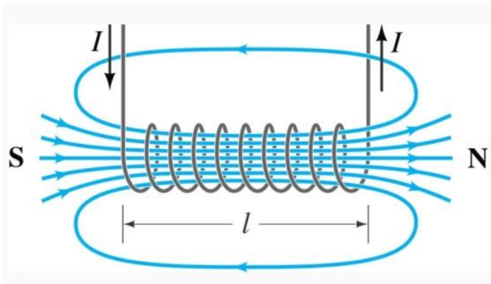
1- The pattern of magnetic flux in a straight wire



2- The magnetic flux in a circular loop



3- The magnetic flux in a solenoid



b- Mention necessary condition for each of the following

1- The magnetic force acting on a straight wire carrying current = zero

The straight wire placed parallel to magnetic field

2- The magnetic torque acting on a rectangular coil carrying current = zero

The coil plane placed perpendicular to magnetic flux

c – The electrical instruments

The instrument	The usage	The idea of operation
1- moving coil galvanometer	measuring and detecting the direction of weak current	The magnetic torque
2- electric motor	Conversion electrical energy into mechanical energy	The magnetic torque

d- State the factor affect for each of the following

1- The magnetic flux density at a point on far normal distance from the wire

The electric current intensity (I) , $B \propto I$ at constant μ , d

The magnetic permeability (μ) , $B \propto \mu$ at constant I , d

The vertical distance between the wire and the point (d) , $B \propto \frac{1}{d}$ at constant B , μ

2- The magnetic flux density at the center of circular loop

The electric current intensity (I) , $B \propto I$ at constant μ , r , N

The magnetic permeability (μ) , $B \propto \mu$ at constant I , N , r

The no.of turns (N) , $B \propto N$ at constant I , r , μ

The radius of the coil (r) $B \propto \frac{1}{r}$ at constant I , μ , N

3- The magnetic flux density at any point along axis of the solenoid

The electric current intensity(I) , $B \propto I$ at constant μ , L, N

The magnetic permeability(μ) , $B \propto \mu$ at constant I , N , L

The no.of turns(N) , $B \propto N$ at constant I , L, μ

The length of solenoid (L) $B \propto \frac{1}{L}$ at constant I , μ , N

4- The magnetic force

The magnetic flux density (B), $F \propto B$ $at\ constant\ I, L, \sin\theta$

The electric current intensity (I), $F \propto I\ at\ constant\ B, L, \sin\theta$

The length of the wire (L) , $F \propto L\ at\ constant\ B, I, \sin\theta$

The sine angle between the wire and the magnetic flux ,

$F \propto \sin\theta\ \ at\ constant\ B, L, I$

5- The magnetic torque

The magnetic flux density (B), $\tau \propto B$ $at\ constant\ I, N, A, \sin\theta$

The electric current intensity (I), $\tau \propto I\ at\ constant\ B, N, A, \sin\theta$

The area electric of the coil(A) , $\tau \propto L\ at\ constant\ B, N, I, \sin\theta$

The no. of turns of the coil (N), $\tau \propto N\ at\ constant\ B, A, I, \sin\theta$

The sine angle between the normal to coil's plane and the magnetic flux $\tau \propto \sin\theta\ at\ constant\ B, A, I, N$

6- The magnetic dipole moment $\left|\overrightarrow{Md}\right|$

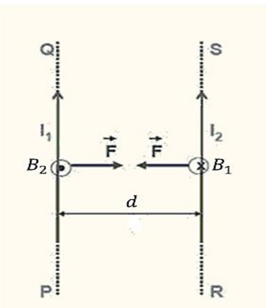
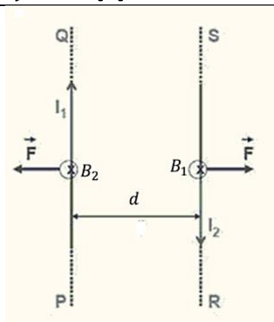
The electric current intensity (I), $\left|\overrightarrow{Md}\right| \propto I\ at\ constant\ A, N$

The no.of turns of the coil (N), $\left|\overrightarrow{Md}\right| \propto N\ at\ constant\ I, N$

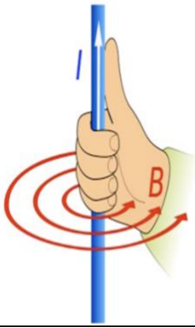
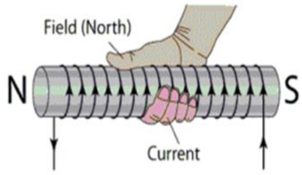
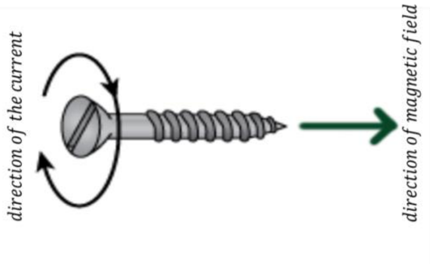
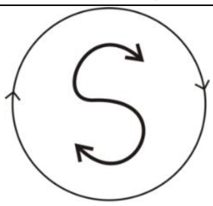
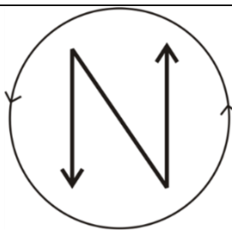
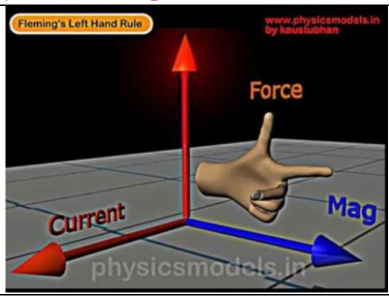
Area of the coil (A), $\left|\overrightarrow{Md}\right| \propto A\ at\ constant\ I, N$

e – Comparisons

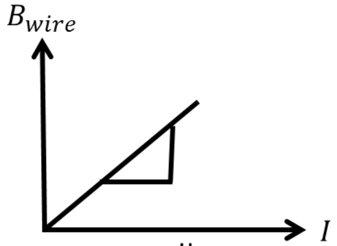
1- Two parallel wires carrying current

Point of comparison	(I) in same direction	(I) in opp. Direction
Direction of magnetic force		
The total magnetic flux (B_t) at any point between two wires	$B_t = B_1 - B_2$	$B_t = B_1 + B_2$
The total magnetic flux density(B_t) at any point outside the two wires	$B_t = B_1 + B_2$	$B_t = B_1 - B_2$
The neutral point	It is appear between them	It is appear outside them

2- The directions

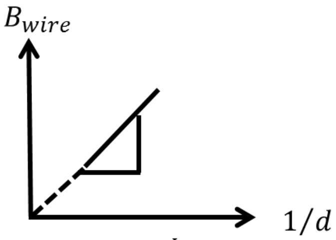
Point of comparison	**a) ampere right hand rule**	
1-The form of illustration		
2- The usage	Determining the direction of magnetic flux density in straight wire	Determining the polarity of magnetic field in a coil or a solenoid
point of comparison	**b) max well right hand screw rule**	
1- The form of illustration		
2- The usage	*Determining the direction of magnetic flux in the center of loop *Direction of magnetic flux density at the axis of solenoid *direction of magnetic dipole moment(Md)	
Point of comparison	**c) Clock wise rule**	
1- the form illustration	 Clockwise Current Polarity - Southpole 	
2- the usage	Determining the type of poles in each of faces loop or solenoid	
Point of comparison	**d) Fleming left hand rule**	
1- The form of illustration		

f- The graphical representation



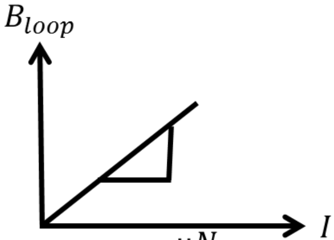
$slope = \frac{\mu}{2\pi d}$

$relation = B = \frac{\mu I}{2\pi d}$



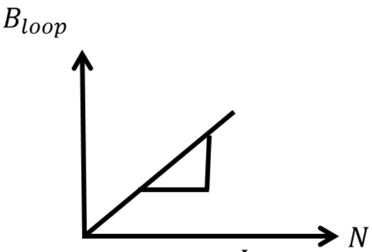
$slope = \frac{\mu I}{2\pi}$

$relation = B = \frac{\mu I}{2\pi d}$



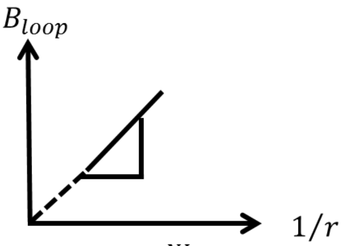
$slope = \frac{\mu N}{2r}$

$relation = B = \frac{\mu NI}{2r}$



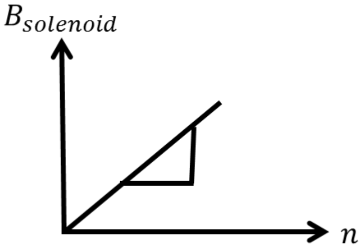
$slope = \frac{\mu I}{2r}$

$relation = B = \frac{\mu NI}{2r}$



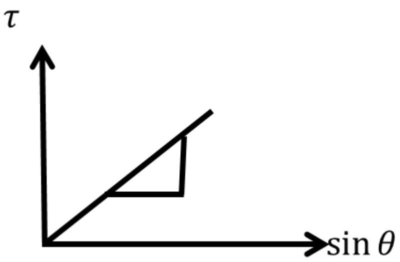
$slope = \frac{\mu NI}{2}$

$relation = B = \frac{\mu NI}{2r}$



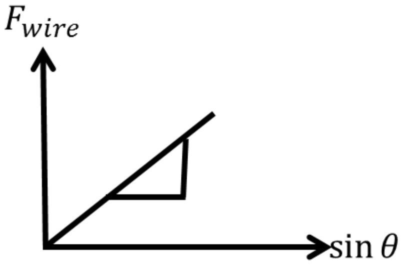
$slope = \mu I$

$relation = B = \mu n I$



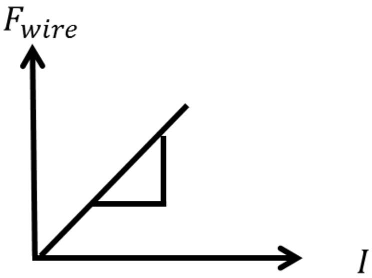
$slope = BIAN$

$relation = \tau = BIAN\sin\theta$



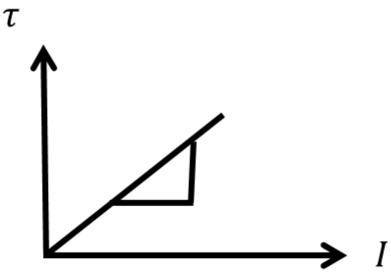
$slope = BIL$

$relation = F = BIL\sin\theta$



$slope = BL\sin\theta$

$relation = F = BIL\sin\theta$

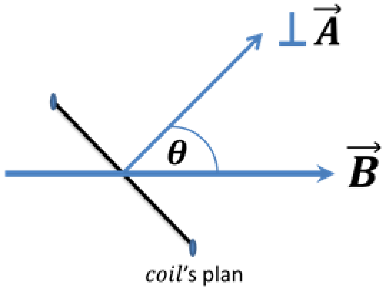


$slope = BAN\sin\theta$

$relation = \tau = BIAN\sin\theta$

f- Relations for chapter two

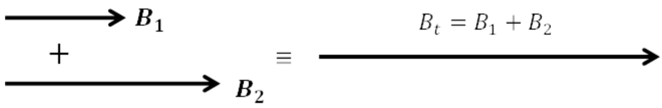
$$* \ \emptyset_m = BA\cos\theta_n$$



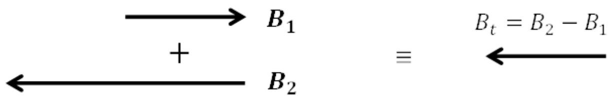
$$* \ B_W = \frac{\mu I}{2\pi d} \qquad * \ B_{center} = \frac{\mu NI}{2r} \qquad * \ B_{solenoid} = \frac{\mu NI}{L_{axis}} = \mu nI$$

$*$ B_t at any point

1- When their fields are the same direction.



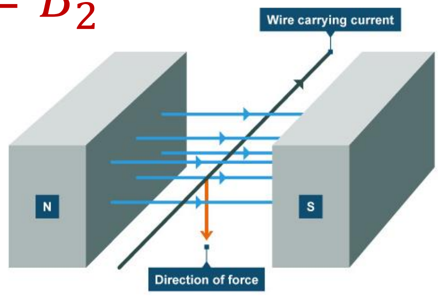
2- When their fields are the same direction.



*** At neutral point**

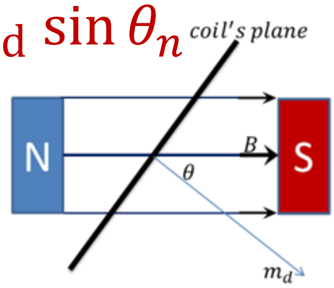
$$B_t = zero \qquad , \qquad B_1 = B_2$$

***the magnetic force** $\quad F = B_{ex} IL \sin\theta$



*** the magnetic torque** $\quad \tau = B_{ex} IAN \sin\theta_n = B_{ex} \, m_d \sin\theta_n$

***the magnetic dipole moment** $\quad m_d = IAN$



Exercise $(\mu_{air} = 4\pi \times 10^{-7} \ Wb/A.m)$

1. A coil of cross-sectional area 0.2 m^2 is placed normal to a uniform magnetic flux of density 0.04 $Weber/m^2$. Calculate the magnetic flux which passes through this coil

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2. A straight wire carrying a current of intensity 4A , if the magnetic density at a point at a certain distance from its axis is $2 \times 10^{-5} T$. Find the distance of the point from the axis of the wire

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3. an electric current passes through a wire of length 26.4 cm bended in a shape of arc of a circle of radius 6.5 cm if the magnetic flux density at the center of the circle $8.25 \times 10^{-6} T$. Calculate the electric current intensity

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4. a solenoid of length 0.6 m is carrying electric current 10 A if the magnetic flux density produced in the middle of its axis is 0.05 T. calculate :

(a) number of turns per unit length

(b) number of turns

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5. a solenoid of length 0.22 m and cross-sectional area $25 \times 10^{-4} m^2$ consists of 300 turns. Calculate the electric current intensity in the coil if the magnetic flux density in the middle of its axis is $1.2 \times 10^{-3}\ Wb/m^2$, then calculate the total magnetic flux passing through the coil

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6. calculate the magnetic force acting on a straight wire of length 50 cm carrying electric current of intensity 2 A placed perpendicular to a magnetic flux of density 0.2 T

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7. a wire carrying electric current of intensity of 10 A is placed perpendicular to a magnetic field of density 0.1 Tesla , calculate the acting force per unit length of the wire

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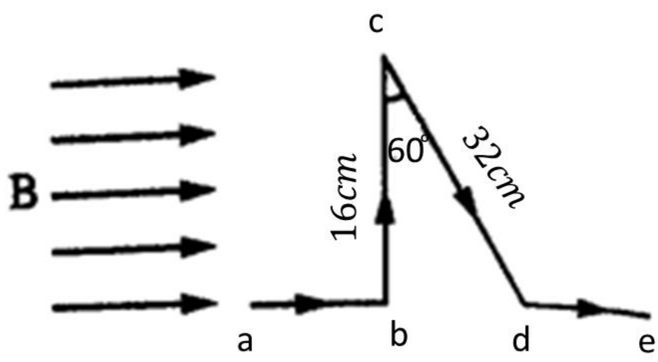
8. in the opposite figure :

If the electric current intensity passing in the wire is 5 A and the flux density is 0.15 T , calculate the force which is acting on parts ab , bc , cd , de



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9. a coil of 500 turns carrying electric current of intensity 10 A is placed in a magnetic field of density 0.25 tesla , if its cross-sectional area is 0.2 m^2 , calculate the torque acting on the coil when the angle between the normal to the coil and the field is 30°

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10. a rectangular coil of dimensions 20 cm , 10 cm and of 200 turns is placed in a uniform magnetic field of magnetic flux density of 0.4 Tesla. A current of 3 Amperes is passed through the coil , calculate the torque acting on the coil in the following cases :

(a) when the coil plane is inclined by angle 60° to the direction of the field

(b) when the coil plane is perpendicular to the direction of the field

(c) when the coil plane is parallel to the direction of the field

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11. a rectangular coil its dimensions 0.12 m and 0.1 m , carries a current 3 A its number of turns 50 turns. The coil is placed perpendicular to a magnetic field whose magnetic flux density uniform. Calculate the magnetic dipole of the coil

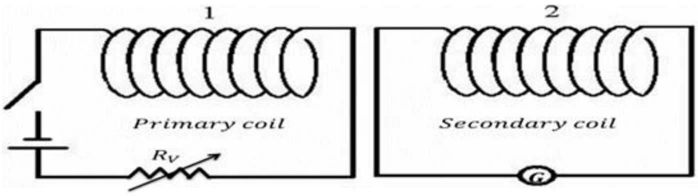
………

Main tablets unit two

a- Mention conditions for each of the following

1- Producing backward induced EMF in a secondary coil

*Close the key in the primary coil
*Adjust the variable resistor to increase the current in the primary coil
*Move the primary coil toward the secondary coil



Mutual induction between 2 coils

2- Producing forward induced EMF in secondary coil

Open the key in the primary coil
Adjust the variable resistor to decrease the current in the primary coil
Move the primary coil away from the secondary coil

b – Some electrical devices based on

The application	The function	idea of operation
Induction furnaces	To melt the metal	Eddy currents
(AC) generator	To convert the kinetic energy into electric energy	Electromagnetic indiction
The electric transformer	Step up or step down voltage	The mutual induction between two coils

c- The directions

1- Fleming right hand rule

Determining the direction of induced current in a straight wire moves ⊥ mag. field

2- Lenz's rule Indicating the direction of induced current in a coil

d – state the factors that effected on

1- The induced EMF in coil

*The rate of variation of a magnetic flux lines $\mathcal{E}_{ind} \propto \frac{\Delta\phi_m}{\Delta t}$
*The no.of turns that cut the magnetic flux $\mathcal{E}_{ind} \propto N$

2- The induced EMF in a straight wire moves in a magnetic field

*The magnetic flux density $\mathcal{E}_{ind} \propto B$
*The length a conductor $\mathcal{E}_{ind} \propto L$ *The velocity of straight wire $\mathcal{E}_{ind} \propto Vel$
*The sine angle between the direction of velocity and the direction of magnetic field

3- The coefficient of mutual induction between two coils

*The magnetic permeability of medium *The volume of coil

*The no.of turns of the coils *The distance between them

4- The coefficient of self-induction

*The geometry *The no.of turns *The length of the coil

5- The instantaneous induced EMF in a dynamos coil

*The no.of turns of the coil $\mathcal{E}_{inds} \propto N$ *The magnetic flux density $\mathcal{E}_{inds} \propto B$

*The area of the coil $\mathcal{E}_{inds} \propto A$ *The angular velocity of the coil $\mathcal{E}_{inds} \propto \omega$

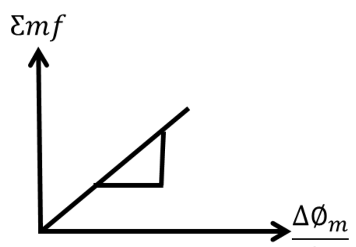
*The sin angle between the magnetic density and the normal to coil $\mathcal{E}_{inds} \propto \sin\theta$

e – Comparisons

2- (Step up, step down) transformer

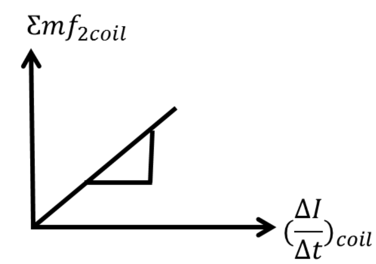
Point of comparison	Step up transformer	Step down transformer
The usage	Step up voltage at electric power station	Step down at electric distribution station
The no.of turns	$N_s > N_p$	$N_s < N_p$
The voltage	$V_s > V_p$	$N_s < N_p$
The electric current	$I_s < I_p$	$I_s > I_p$

f – The graphical representation



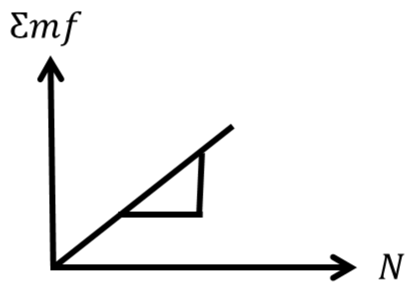
$slope = emf = -N\frac{\Delta \phi_m}{\Delta t}$

$relation = N$



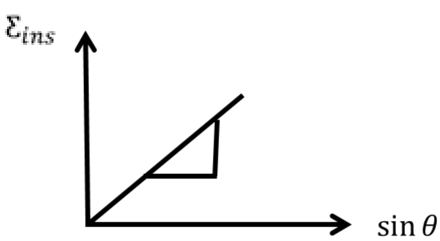
$slope = \mu$

$relation = \mathcal{E}mf = -\mu\frac{\Delta I}{\Delta t}$



$slope = BN\omega$

$relation = \mathcal{E}_{max} = BAN\omega$

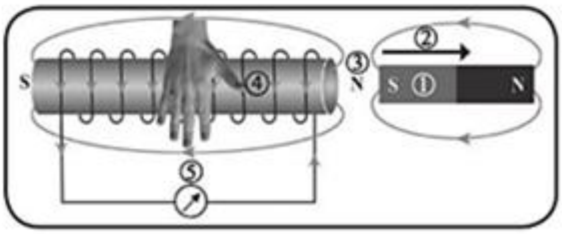


$slope = ABN2\pi F = \mathcal{E}_{max}$

$relation = \mathcal{E}_{ins} = ABN2\pi F\sin\theta$

f- Rules

➢ Faraday and Straight Wire:



$$emf_{av} = -N\frac{\Delta\phi_m}{\Delta t} \qquad emf = \frac{\Delta (BA \cos\theta)}{\Delta t}$$

Force needed to move the wire with constant speed: $F = BIL$

➢ Mutual and Self-induction:

1- $emf_2 = -M\frac{\Delta I_1}{\Delta t} = -N_2\frac{\Delta \phi_2}{\Delta t}$

2- $M \times \Delta I_1 = N_2 \times \Delta\phi_2$

3- $emf = -L\frac{\Delta I}{\Delta t} = -N\frac{\Delta \phi}{\Delta t}$

4- $L \times \Delta I = N \times \Delta\phi$

➢ Dynamo:

1- $\Sigma_{inst} = \Sigma_{max}\sin\theta = ABN\omega \sin\theta = ABN2\pi F \sin\theta$

2- $I_{inst} = I_{max}\sin\theta$ 3- $\Sigma_{inst} = I_{inst} \times R_{coil}$

When the coil starts from zero position (no. of times)

$N_{max/sec} = 2f$ $N_{zero/sec} = 2f + 1$

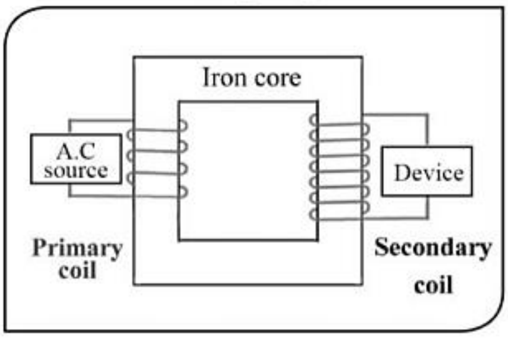
➢ Transformer.

$$\frac{\eta V_P}{V_S} = \frac{I_S}{I_P} = \frac{N_P}{N_S}$$

$\eta P_P = P_S$

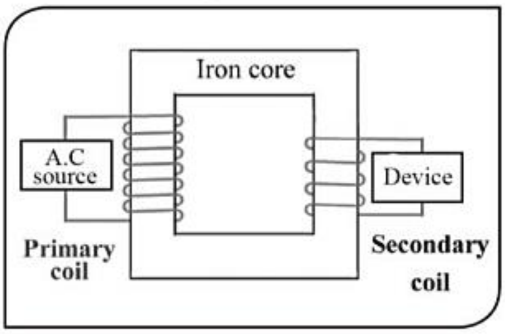
$\eta = 100\% = 1$ (ideal)

Step-up



$N_P < N_S$
$V_P < V_S$
$I_P > I_S$

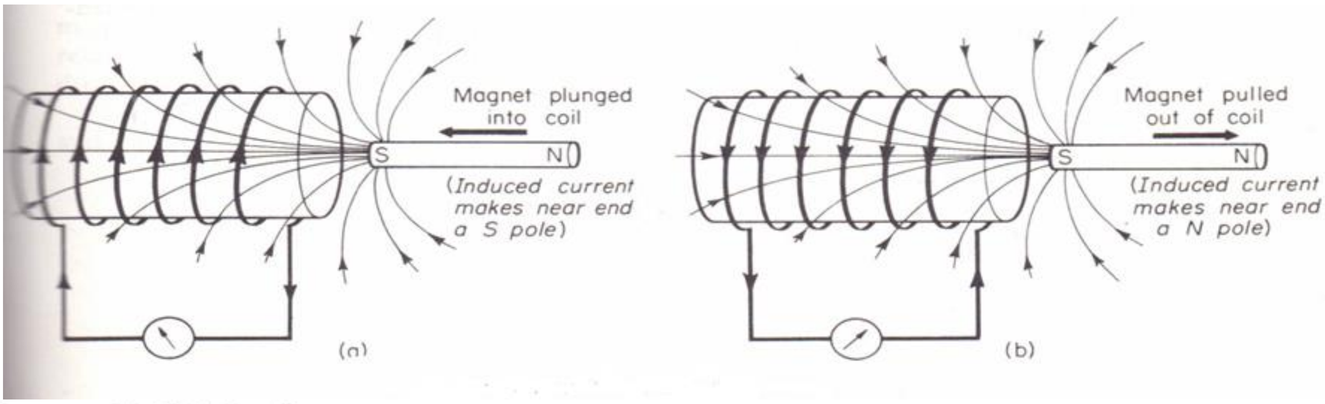
Step-down



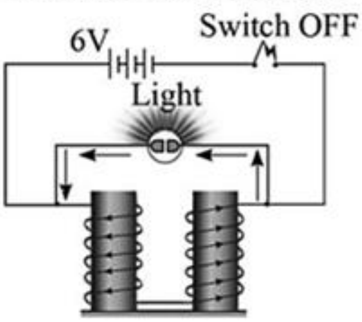
$N_P > N_S$
$V_P > V_S$
$I_P < I_S$

g - Experiments

1- Faraday's experiment



In case of self induction:



Switch (ON)

The lamp will not Glow (Low battery voltage)

Switch (OFF)

Lamp Glow (Large forward induced emf) and spark passed between two terminals of the switch (due to large number of turns)

Exercise

1- A solenoid coil of 200 turns, the cross – sectional area of each turn equals $2cm^2$, it is placed normally to a magnetic field of flux density 0.6 Wb/m^2. Calculate the induced emf for the following cases when :

a) Flux density increases to 0.8 Tesla in $2 \times 10^{-3} sec$

b) The coil is turned back (rotates 180^0) in 0.1 sec

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2- If the coefficient of mutual induction between two adjacent coils is 0.2 Henry and the electric current intensity in one of them is 4 A, then it vanished within 0.01 s , calculate the average induced emf generated in the secondary coil

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3- Two adjacent coils A, B their number of turns are 100 , 200 turns respectively , if an electric current of 2 A passed in coil A it generates a magnetic flux of $3 \times 10^{-4} Wb$ and generates $1.5 \times 10^{-4} Wb$ in coil B. find :

(b) Mutual induction coefficient between the two coils

(c) Average emf in coil B when the electric current vanishes in coil A within 0.1 s

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4- Two adjacent coils A , B the number of their turns 500 , 2000 turn respectively. When the electric current in A is changes by 10 A , the magnetic flux in coil A changed by $2 \times 10^{-3} Wb$

and in a coil B changed by $10^{-4}\ Wb$, find :(a) coefficient of self induction of coil A

(b) the average generated in each of the two coils A , B if an electric current of 15 A passing in coil A is cut within a time of 0.3 s

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5- The coil of a simple AC generator consists of 100 turn , the cross – sectional area of each is 0.21 m^2. The coil rotates with frequency 50 Hz (cycles/second) in a magnetic field of constant flux density $B = 10^{-3}\ Weber/m^2$. What is the maximum induced emf generated ?And what is its value when the angle between the direction of velocity and flux direction is $\theta = 30^{\circ}$?

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6- Step down transformer of efficiency 100% is required to be used to operate an electric lamp of power 24W and potential difference 12V using a power source of emf 240V if the number of turns of the secondary coil is 480 calculate :

(a) The electric current intensity passing through the secondary coil

(b) The number of turns of the primary coil

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7- Step down transformer drops down the electric potential from 2400V to 120V and the number of turns of its primary coil is 4000 turn , if it delivers an electric power of 13500W and its efficiency is 90% , find :

(a) The number of turns of the secondary coil

(b) The electric current intensity in the two coils

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Main tablets unit three

- ## Part one: Dual nature

a - Conditions of occurrence of some physical processes

1- Emission of electrons from a metallic surface

Electrons gain sufficient energy (thermal or light energy) greater than the work function of the metallic surface.

2- Emission of photoelectric electrons from a metallic surface

Falling light of frequency is greater than the critical frequency of the metallic surface.

b- Mention one application or use for each of the following

a- Remote sensing devices (thermal photography- infrared rays).

1-in military (night vision system.) 2- in medicine (tomography – embryology)

3- in criminology 4- Indicating the mineral earth resources.

b- Microwaves devices. In radar.(Indicating the mineral earth resources)

c- Cathode rays tube. Computer monitor – screen of T.V

d- Photoelectric cell Changing the light energy into electric energy (in calculators)

i- Grid in cathode ray tube Controlling in the intensity of electrons beam

j- Raster. Controlling in the deviation of electron beam to fill the screen

k- The tube in CRT is evacuated. To move the liberated electron beam easily

l- Anode in CRT. To accelerate the electron beam

m- Wine's law. Indicating the temperature of stars

c- Mention the scientific idea of operation for

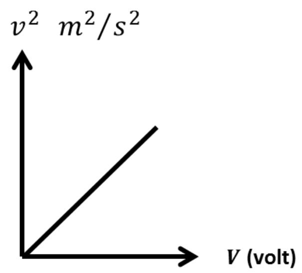
2- Cathode rays tube: thermionic effect

3- Photoelectric cell: photoelectric effect

5- Atomic bomb: Einstein's relation $E = mc^2$

E- Graphical representation:

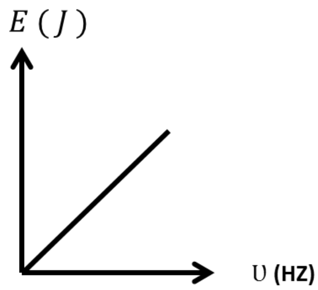
1 –



$relation : eV = \frac{1}{2}mv^2$

$slope = \frac{\Delta v^2}{\Delta V} = \frac{2e}{m_e}$

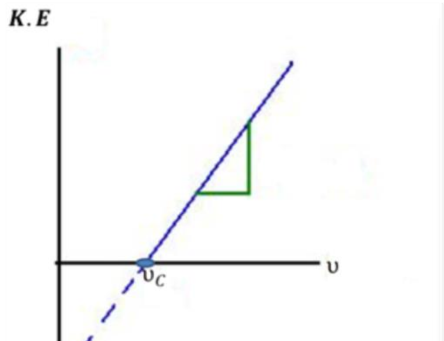
2 –



$relation : h\,\upsilon$

$slope = \frac{\Delta E}{\Delta \upsilon} = h$

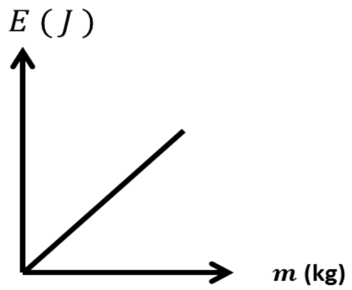
3 –



$relation : K.E = h(\upsilon - \upsilon_c)$

$slope = \frac{\Delta K.E}{\Delta(\upsilon - \upsilon_c)} = h$

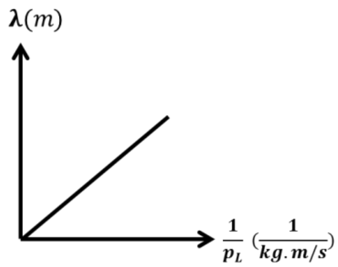
4 –



$relation : E = mc^2$

$slope = \frac{\Delta E}{\Delta m} = c^2$

7 –



$relation : \lambda = \frac{h}{p_L}$

$slope = \frac{\Delta\lambda}{\Delta\frac{1}{P_L}} = h$

E- Rules:

. Black body radiation

Wien's law:

$$\frac{\lambda_{max\,(1)}}{\lambda_{max\,(2)}} = \frac{T_2}{T_1} \qquad T(Kelvin) = t\,(Celisus) + 273$$

. Cathode ray tube(based on thermionic effect)

$$1 - K.E = \frac{1}{2} m_e v^2 = e\,V\ , \quad 2 - \ v = \sqrt{\frac{2K.E}{m_e}} = \sqrt{\frac{2\,e\,V}{m_e}}$$

$$K.E.\,(eV) \xrightarrow{\times(1.6\times10^{-19})} K.E.\,(Joule) \qquad K.E.\,(Joule) \xrightarrow{\div(1.6\times10^{-19})} K.E.\,(eV)$$

.photoelectric effect

- $k.E = E_{photon} - E_{work}$

$$1\text{- } K.E = \frac{1}{2} m_e v^2 = e\,V, \quad 2\text{- } E_{photon} = h\,\upsilon_{photon} = \frac{hc}{\lambda_{photon}}, \quad 3\text{- } E_w = h\,\upsilon_c = \frac{hc}{\lambda_c}$$

-Compton effect

1- summation $(E_{photon} + K.E_{electron})$ Before the collision = summation $(E_{photon} + K.E_{electron})$ After the collision

Another way:

ΔE Energy loss of photon = $\Delta K.E$ Energy gain of electron (electron ال بيكتسبه energyال من photon ا اللى بيخسروا)

.properties of photon:

- ✓ It has a momentum: $P_L = m \times c = \frac{E}{C} = \frac{h\upsilon}{c} = \frac{h}{\lambda}$
- ✓ The energy of photon: $E = mc^2 = h\,\nu = \frac{hc}{\lambda}$

. electron De Broglie states that:

- ✓ K.E $_{elctrons} = e\,V = \frac{1}{2} m_e v^2$ *so when* Voltage ↑ , K.E elctron ↑ velocity ↑
- ✓ It has a momentum: $P_L = m \times v$

Exercise

Planck's constant (h) = 6.625×10^{-34} j.s , velocity of light (C) = $3\times 10^{8}\ m/s$, electron charge (e)= 1.6×10^{-19} C and mass of electron (m) = 9.1×10^{-31} Kg

1) If the temperature of surface of sun is $6000^{\circ}K$ and wave length (λ_m) at maximum radiation intensity is 0.499 nm. Calculate the temperature of filament lamp that give photons at wavelength (λ_m) = 1.2 nm

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2) The energy of photon 3.3$\times 10^{-19}$joule. Calculate its mass , frequency in space and wavelength then state its mass when its velocity is equal zero

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3) When monochromatic light of wavelength $5000^{\circ}A$ incident on a metal surface , electrons of velocity 2.57$\times 10^{5}\ m/s$ is released. Calculate the work function of metal. Do electrons release from this surface when other monochromatic light of wavelength $6000^{\circ}A$ incident on it ?

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14) In an experiment based on Compton effect , a photon of frequency $10^{15}Hz$ collides with an electron at rest. If the scattered photon frequency$5\times 10^{14}Hz$. Find the increase in energy of electron after collision

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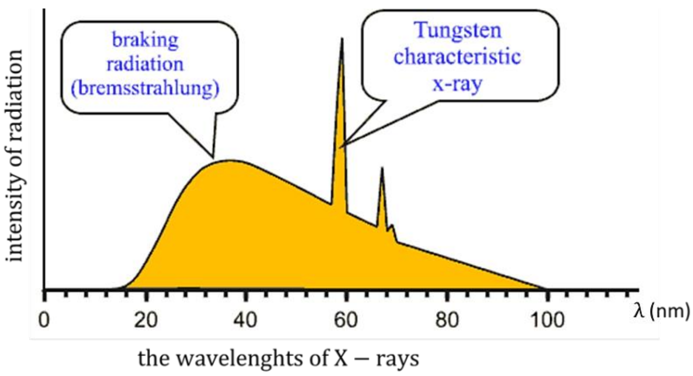
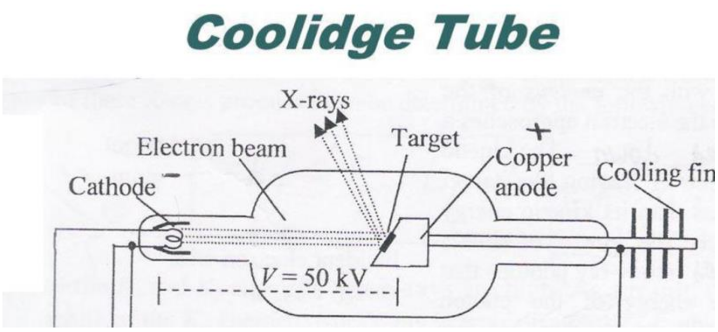
- **Part two: x rays**





a. Uses

x-rays	1. studying the crystalline structure of materials 2. detecting the defects of the used materials in metallic industries 3. photography of the bones and detection of its broken places and fracture in some medical diagnoses
Filament in Coolidge tube	Electrons source
High direct potential difference between cathode and target in Coolidge tube	Accelerating emitted electrons from cathode (filament)

b.comparison

1- Point of comparison	***Continuous spectrum***	***Line spectrum***
Adjectives	(braking or soft) radiation.	(hard) radiation.
The Factor affecting	*The voltage between the cathode and the target material	*The type of target element ,it must have high atomic number
Source of radiation	The liberated electrons from the cathode	The electrons of target material

Rules

1- **continuous radiation of x rays** $eV = h\frac{C}{\lambda_{min}} = h\upsilon_{max} = E_{max}$

2- **characteristic x rays** $\Delta E_{target} = E_{x\text{-ray line}} = h\upsilon_{line} = \frac{hc}{\lambda_{line}}$

Exercise

Planck's constant (h) = 6.625 $\times 10^{-34}$j.s , velocity of light (C) = 3$\times 10^{8}\ m/s$, electron charge (e)= 1.6$\times 10^{-19}$C and mass of electron (m) = 9.1$\times 10^{-31}$Kg

1) an x-ray tube is operating at 100 KV. calculate the minimum wavelength of x-rays and maximum velocity of the electrons striking the anode

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2) If you know that the shortest wavelength of the produced X-rays from Coolidge tube is 0.414 Å, calculate:

(i) The maximum energy for the photons of X-rays equals

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(ii) The applied potential difference between cathode and anode equals

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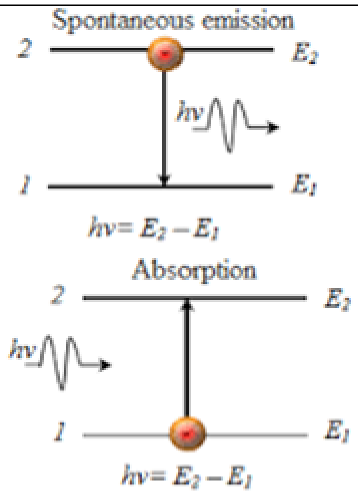
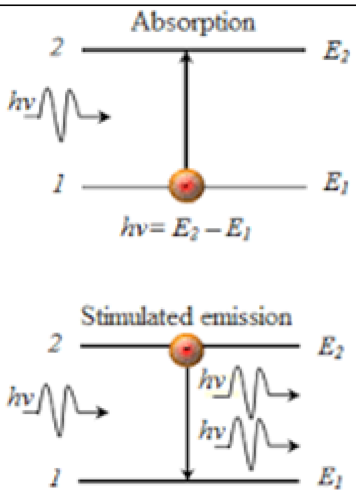
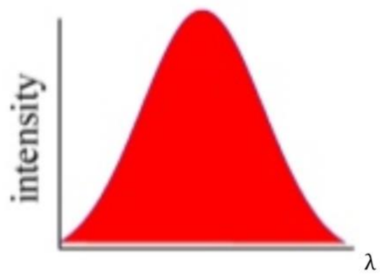
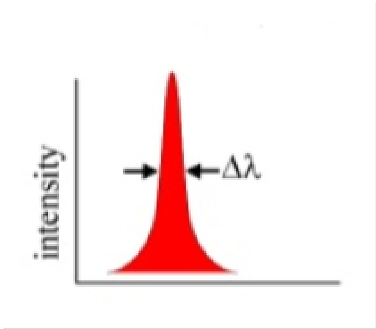
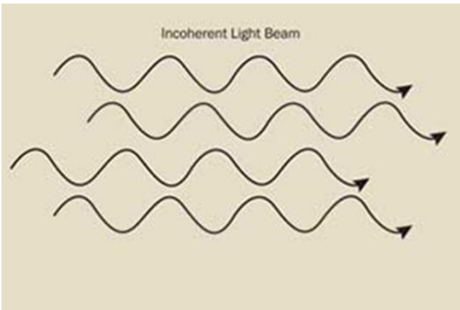
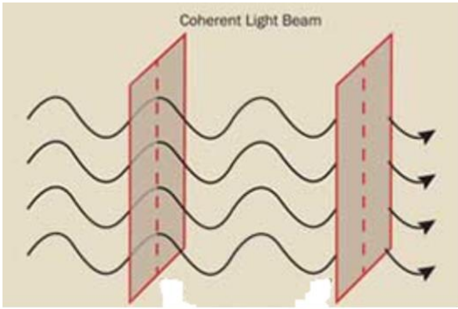
• Part three: laser

c- Comparison

1-Point of comparison	Spontaneous emission	Stimulated emission
Representing drawing		
Occurrence	When the atom relaxes from higher energy level (excited state) to a lower state after the lifetime is over. A photon emits spontaneously, its energy equals the difference in energy levels.	When an external photon stimulates an excited atom to relax from the excited state before the lifetime is over.

2- Point of comparison	Ordinary light sources	Laser sources
Monochromaticity	 (different shades)	 (one wavelength).
Collimation	The diameter increases with the distance due to the scattering.	The diameter stays constant for long distances
Coherence		
Intensity	decreases with distance increase	constant

✓ **Inverse square low** the intensity of light α $(\frac{1}{distance})^2$ $\frac{I_1}{I_2} = \frac{d_2^2}{d_1^2}$

Exercise

a. choose the correct answer

1) Laser photons consider as :

(emission line spectru – absorption line spectrum – continuous spectrum)

2) The velocity of laser beam is …. The velocity of ordinary light

(higher than – equal to – lower than)

3) Laser beam is highly intensity , which means........................

(it does not obey inverse square law – it has only one wavelength – its photons have the same direction)

4) Laser beam does not follow inverse square law because of ....

(its monochromaticity - its coherent – its non divergence)

5) Laser beam differs from light ray in.........................

(the coherency of its photon – obeying the inverse square law – the color of the ray)

6) The lifetime of an atom in a metastable state is of the order of ....

$$(10^{-8}s - 10^{-10}s - 10^{-3}s - 10^{-1}s)$$

7) After optical pumping process in laser....................

(the number of atoms in the ground state is greater than the number of atoms in the excited state – the number of atoms in the excited state is greater than the number of atoms in the ground state – the number of atoms in the ground state is equal to the number atoms in the excited state – no atoms are available in the excited state)

8) The emission from neon lamp is..........

(spontaneous – stimulated – absorption)

11) Through holography we can produce ………. Dimensional images of objects

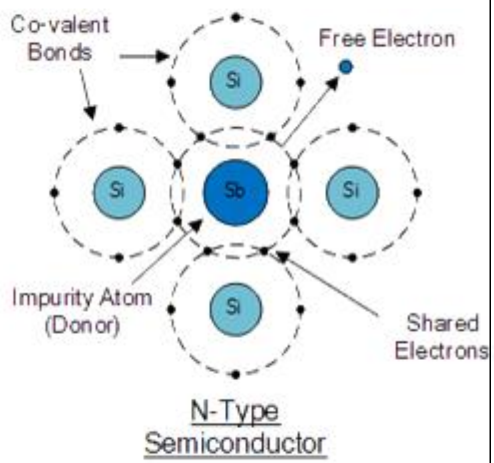
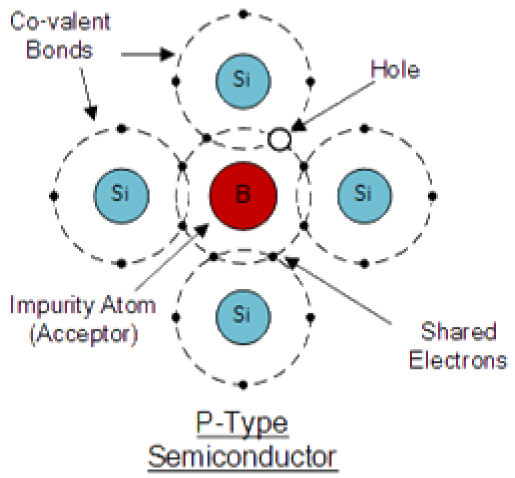
(one – two – three)

Main tablets unit four

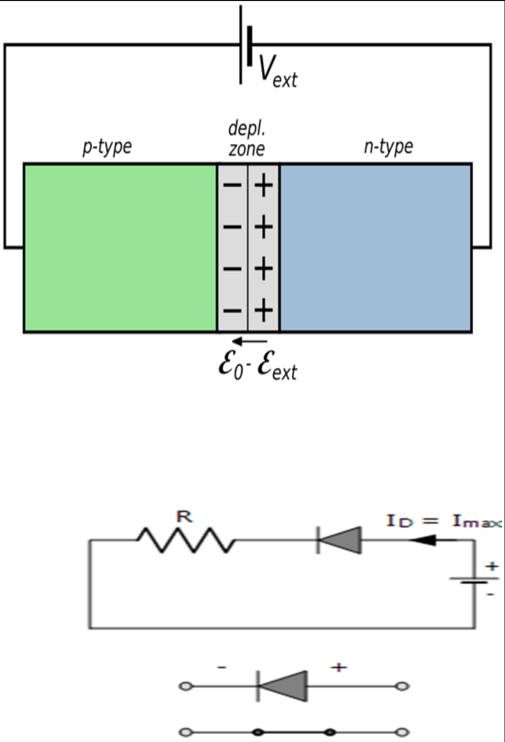
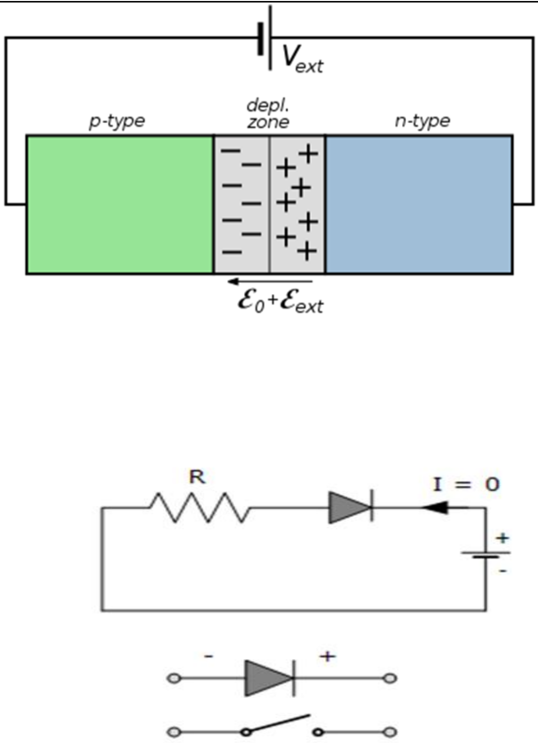
a- Uses

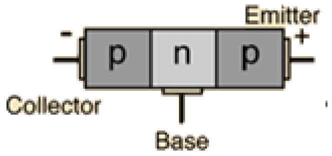
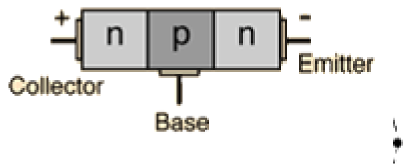
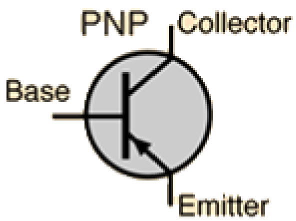
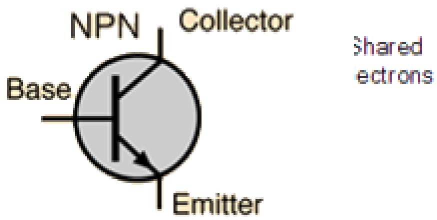
1- diode	1-acts as opened key (backward connection) -acts as closed key (forward connection) 2- rectifying AC to DC to charge cars batteries and mobiles chargers
2- transistor	1- as amplifier
3- Analog devices	1- microphone 2- video camera 3- normal television
4- digital devices	1- cell phones (mobiles) 2- CDs 3- computers 4- digital channel

b- Comparisons

1- P.O.C	Donors impurities	Acceptors impurities
Shape of crystal after doping		
Impurity valence	Pentavalent atoms like phosphorus (p) and antimony (Sb)	Trivalent atom as boron (B) and aluminum (AL)
Charges carriers	Free electrons	Holes

Impurity atom after doping	Became positive ions N_D^+	Became negative ions N_A^-
Crystal type after doping	n-type	p-type
Relation bet n,p	$N > P$	$N < P$

2- P.O.C	**Forward bias of pn junction**	**Reversed bias of pn junctions**
Method of connections		
Effect the voltage of the battery	The external field (battery) is the **opposite direction** to the internal field	The external field (battery) is **the same direction** to the internal field)
Voltage of the pn junction	The voltage of **pn** junction is less that the barrier voltage	The voltage of **pn** junction is greater than the barriers voltage
current	High	Zero

3- P.O.C	p n p transistors	n p n transistors
Structure		
Symbol in electric circuit		

<table>
<tr><th>4- P.O.C</th><th>Inversion (NOT)</th><th>Coincidence(AND)</th><th>Optionality(OR)</th></tr>
<tr><td>Number of input and output :</td><td>One input and one output</td><td>Two or more inputs and one output</td><td>Two or more inputs and one output</td></tr>
<tr><td>Logic operation :</td><td>Inverse (the output is inverse of the input)</td><td>Simultaneity the output is not one unless all inputs are one</td><td>Optional the output is one if one or more of inputs is one</td></tr>
<tr><td>Truth table</td>
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<td><table><tr><td>Input</td><td>Input</td><td rowspan="2">Output</td></tr><tr><td>A</td><td>B</td></tr><tr><td>0
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<td><table><tr><td>input</td><td>input</td><td rowspan="2">Output</td></tr><tr><td>A</td><td>B</td></tr><tr><td>0
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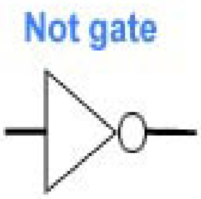
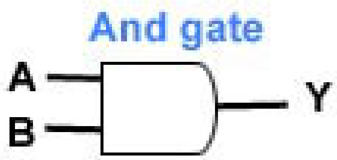
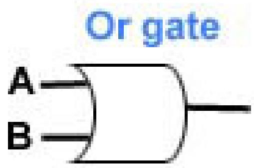
<td></td>
<td></td>
<td></td></tr>
</table>

NAND Gate

It is an AND gate followed by NOT gate, has two or more inputs with one output

The electronic circuit that gives a low output (0) only if **all** its inputs are high

symbol

Truth table

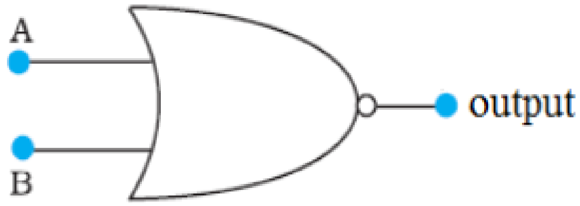
A	B	Output
0	0	1
0	1	1
1	0	1
1	1	0

NOR Gate

It is an OR gate followed by NOT gate, has two or more inputs with one output

The electronic circuit that gives a high output (1) only if **all** its inputs are low

symbol



Truth table

A	B	Output
0	0	1
0	1	0
1	0	0
1	1	0

c.Rules

$$\propto_e = \frac{I_C}{I_E} = \frac{\beta_e}{(1+\beta_e)} \qquad \beta_e = \frac{I_C}{I_B} = \frac{\propto_e}{(1-\propto_e)}$$

Exercise

1) Collector current of transistor is 10 mA and base current 100 μA. calculate the current division "$\propto_e$" and the current gain "β_e"

....

....

2) A transistor has $\beta_e = 24$ calculate $\propto_e$ then calculate the collector current if base current is 24 μA

....

....

3) The opposite electronic circuit represents combining gates for a special job. Complete the following truth table for this circuit:

A	B	Output
0	0	
1	0	
0	1	
1	1